

● HARD

● GEOMETRY AND TRIGONOMETRY

● ANSWER KEY

Area and volume

03

9 answers · Rationale-rich · Teach from doc

Q1**36504**

The correct answer is 36,504. The volume of a cube can be found using the formula $V = s^3$, where s represents the edge length of a cube. It's given that this cube has a volume of 474,552 cubic units. Substituting 474,552 for V in $V = s^3$ yields $474,552 = s^3$. Taking the cube root of both sides of this equation yields $78 = s$. Thus, the edge length of the cube is 78 units. Since each face of a cube is a square, it follows that each face has an edge length of 78 units. The area of a square can be found using the formula $A = s^2$. Substituting 78 for s in this formula yields $A = 78^2$, or $A = 6,084$. Therefore, the area of one face of this cube is 6,084 square units. Since a cube has 6 faces, the surface area, in square units, of this cube is 66,084, or 36,504.

Q2**D****D. 41,200**

Choice D is correct. The volume, V , of a sphere is given by $V = \frac{4}{3}\pi r^3$, where r is the radius of the sphere. Since a hemisphere is half of a sphere, it follows that the volume, V , of a hemisphere is given by $V = \frac{14}{23}\pi r^3$, or $V = \frac{2}{3}\pi r^3$. Substituting 27 for r in this formula yields $V = \frac{2}{3}\pi 27^3$, which gives $V = 13,122\pi$, or V is approximately equal to 41,223.98. Therefore, the choice that is closest to the volume, in cubic inches, of this hemisphere is 41,200.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q3

A

A. $\frac{1}{3}$

Choice A is correct. The equation for the volume of a right circular cone is $V = \frac{1}{3} \pi r^2 h$. It's given that the volume of the right circular cone is $\frac{1}{3} \pi$ cubic feet and the height is 9 feet.

Substituting these values for V and h, respectively, gives $\frac{1}{3} \pi = \frac{1}{3} \pi r^2 (9)$. Dividing both sides of the equation by $\frac{1}{3} \pi$ gives $1 = r^2 (9)$. Dividing both sides of the equation by 9 gives $\frac{1}{9} = r^2$.

Taking the square root of both sides results in two possible values for the radius, $\sqrt{\left(\frac{1}{9}\right)}$ or $-\sqrt{\left(\frac{1}{9}\right)}$. Since the radius can't have a negative value, that leaves $\sqrt{\left(\frac{1}{9}\right)}$ as the only

possibility. Applying the quotient property of square roots, $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$, results in $r = \frac{\sqrt{1}}{\sqrt{9}}$, or $r = \frac{1}{3}$.

Choices B and C are incorrect and may result from incorrectly evaluating $\sqrt{\left(\frac{1}{9}\right)}$. Choice D is incorrect and may result from solving $r^2 = 9$ instead of $r^2 = \frac{1}{9}$.

Q4**C****C.** 600π

Choice C is correct. The volume, V , of a right circular cylinder is given by the formula $V = \pi r^2 h$, where r is the radius of the base of the cylinder and h is the height of the cylinder. It's given that a right circular cylinder has a height of 6 meters. Therefore, $h = 6$. It's also given that the right circular cylinder has a base with a circumference of 20π meters. The circumference, C , of a circle is given by $C = 2\pi r$, where r is the radius of the circle. Substituting 20π for C in the formula $C = 2\pi r$ yields $20\pi = 2\pi r$. Dividing each side of this equation by 2π yields $10 = r$. Substituting 10 for r and 6 for h in the formula $V = \pi r^2 h$ yields $V = \pi (10)^2 6$, or $V = 600\pi$. Therefore, the volume, in cubic meters, of the cylinder is 600π .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the lateral surface area, not the volume, of the cylinder.

Choice D is incorrect. This is the result of using the diameter, not the radius, for the value of r in the formula $V = \pi r^2 h$.

Q5**27556**

The correct answer is 27,556. The area of a square is s^2 , where s is the side length of the square. Let x represent the length of each side of square B. Substituting x for s in s^2 yields x^2 . It follows that the area of square B is x^2 . It's given that square A has side lengths that are 166 times the side lengths of square B. Since x represents the length of each side of square B, the length of each side of square A can be represented by the expression $166x$. It follows that the area of square A is $166x^2$, or $27,556x^2$. It's given that the area of square A is k times the area of square B. Since the area of square A is equal to $27,556x^2$, and the area of square B is equal to x^2 , an equation representing the given statement is $27,556x^2 = kx^2$. Since x represents the length of each side of square B, the value of x must be positive. Therefore, the value of x^2 is also positive, so it does not equal 0. Dividing by x^2 on both sides of the equation $27,556x^2 = kx^2$ yields $27,556 = k$. Therefore, the value of k is 27,556.

Q6**24.5**Accept also: $49/2$

The correct answer is 24.5. It's given that a triangle is formed by connecting the three points shown, which are $-3, 4$, $5, 3$, and $4, -3$. Let this triangle be triangle A. The area of triangle A can be found by calculating the area of the rectangle that circumscribes it and subtracting the areas of the three triangles that are inside the rectangle but outside triangle A. The rectangle formed by the points $-3, 4$, $5, 4$, $5, -3$, and $-3, -3$ circumscribes triangle A. The width, in units, of this rectangle can be found by calculating the distance between the points $5, 4$ and $5, -3$. This distance is $4 - -3$, or 7. The length, in units, of this rectangle can be found by calculating the distance between the points $5, 4$ and $-3, 4$. This distance is $5 - -3$, or 8. It follows that the area, in square units, of the rectangle is 78, or 56. One of the triangles that lies inside the rectangle but outside triangle A is formed by the points $-3, 4$, $5, 4$, and $5, 3$. The length, in units, of a base of this triangle can be found by calculating the distance between the points $5, 4$ and $5, 3$. This distance is $4 - 3$, or 1. The corresponding height, in units, of this triangle can be found by calculating the distance between the points $5, 4$ and $-3, 4$. This distance is $5 - -3$, or 8. It follows that the area, in square units, of this triangle is $\frac{1}{2}81$, or 4. A second triangle that lies inside the rectangle but outside triangle A is formed by the points $4, -3$, $5, 3$, and $5, -3$. The length, in units, of a base of this triangle can be found by calculating the distance between the points $5, 3$ and $5, -3$. This distance is $3 - -3$, or 6. The corresponding height, in units, of this triangle can be found by calculating the distance between the points $5, -3$ and $4, -3$. This distance is $5 - 4$, or 1. It follows that the area, in square units, of this triangle is $\frac{1}{2}16$, or 3. The third triangle that lies inside the rectangle but outside triangle A is formed by the points $-3, 4$, $-3, -3$, and $4, -3$. The length, in units, of a base of this triangle can be found by calculating the distance between the points $4, -3$ and $-3, -3$. This distance is $4 - -3$, or 7. The corresponding height, in units, of this triangle can be found by calculating the distance between the points $-3, 4$ and $-3, -3$. This distance is $4 - -3$, or 7. It follows that the area, in square units, of this triangle is $\frac{1}{2}77$, or 24.5. Thus, the area, in square units, of the triangle formed by connecting the three points shown is $56 - 4 - 3 - 24.5$, or 24.5. Note that 24.5 and $49/2$ are examples of ways to enter a correct answer.

B. 8

Choice B is correct. Let x represent the side length, in cm, of each square base. If the two prisms are glued together along a square base, the resulting prism has a surface area equal to twice the surface area of one of the prisms, minus the area of the two square bases that are being glued together, which yields $2K - 2x^2$ cm². It's given that this resulting surface area is equal to $\frac{92}{47}K$ cm², so $2K - 2x^2 = \frac{92}{47}K$. Subtracting $\frac{92}{47}K$ from both sides of this equation yields $2K - \frac{92}{47}K - 2x^2 = 0$. This equation can be rewritten by multiplying $2K$ on the left-hand side by $\frac{47}{47}$, which yields $\frac{94}{47}K - \frac{92}{47}K - 2x^2 = 0$, or $\frac{2}{47}K - 2x^2 = 0$. Adding $2x^2$ to both sides of this equation yields $\frac{2}{47}K = 2x^2$. Multiplying both sides of this equation by $\frac{47}{2}$ yields $K = 47x^2$. The surface area K , in cm², of each rectangular prism is equivalent to the sum of the areas of the two square bases and the areas of the four lateral faces. Since the height of each rectangular prism is 90 cm and the side length of each square base is x cm, it follows that the area of each square base is x^2 cm² and the area of each lateral face is $90x$ cm². Therefore, the surface area of each rectangular prism can be represented by the expression $2x^2 + 490x$, or $2x^2 + 360x$. Substituting this expression for K in the equation $K = 47x^2$ yields $2x^2 + 360x = 47x^2$. Subtracting $2x^2$ and $360x$ from both sides of this equation yields $0 = 45x^2 - 360x$. Factoring x from the right-hand side of this equation yields $0 = x(45x - 360)$. Applying the zero product property, it follows that $x = 0$ and $45x - 360 = 0$. Adding 360 to both sides of the equation $45x - 360 = 0$ yields $45x = 360$. Dividing both sides of this equation by 45 yields $x = 8$. Since a side length of a rectangular prism can't be 0, the length of each square base is 8 cm.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q8**D**

D. 1,849

Choice D is correct. The area of a circle can be found by using the formula $A = \pi r^2$, where A is the area and r is the radius of the circle. It's given that the radius of circle A is $3n$. Substituting this value for r into the formula $A = \pi r^2$ gives $A = \pi 3n^2$, or $9\pi n^2$. It's also given that the radius of circle B is $129n$. Substituting this value for r into the formula $A = \pi r^2$ gives $A = \pi 129n^2$, or $16,641\pi n^2$. Dividing the area of circle B by the area of circle A gives $\frac{16,641\pi n^2}{9\pi n^2}$, which simplifies to 1,849. Therefore, the area of circle B is 1,849 times the area of circle A .

Choice A is incorrect. This is how many times greater the radius of circle B is than the radius of circle A .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the coefficient on the term that describes the radius of circle B .

Q9

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Answer not provided in source data — see rationale below.

Choice D is correct. The surface area is found by summing the area of each face. A right rectangular prism consists of three pairs of congruent rectangles, so the surface area is found by multiplying the areas of three adjacent rectangles by 2 and adding these products. For this prism, the surface area is equal to $2(4 \cdot 5) + 2(5 \cdot 6) + 2(4 \cdot 6)$, or $2(20) + 2(30) + 2(24)$, which is equal to 148.

Choice A is incorrect. This is the area of one of the faces of the prism. Choice B is incorrect and may result from adding the areas of three adjacent rectangles without multiplying by 2. Choice C is incorrect. This is the volume, in cubic inches, of the prism.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

Answers recovered from rationale for Q9 — the source CB data omitted a structured key for these items, so the answer slot shows —. The worked solution in the rationale below each entry contains the correct value.

